Guide Questions:

1. What is the Internet of Things (IoT)?

What is IoT and how does it work?

The Internet of Things (IoT) is a system where physical devices are connected to the Internet and can share information. Devices use sensors to collect data, send it through a network, and use the data to perform an action or give information to the user.

What are the main components of an IoT system?

The main parts are sensors, devices, network connection, data processing, and software. Sensors collect information, the network sends the data, and software processes the information and helps control the device.

Why is IoT important today?

IoT is important because it makes many tasks easier and faster. It can help people monitor devices, automate tasks, save time, and make better decisions using collected information.

2. IoT vs. Internet vs. Embedded Systems

What is the difference between the Internet, Embedded Systems, and IoT?

The Internet is a network that allows devices to communicate and share information. An embedded system is a small computer inside a device that performs a specific task. IoT connects physical devices to a network so they can collect and exchange data.

Can an embedded system exist without the Internet? Can IoT exist without embedded systems?

Yes, an embedded system can work without the Internet. For example, a digital clock can work without being connected online. IoT usually needs embedded computing systems because devices need something to collect, process, or send data. However, not all embedded systems are IoT devices.

Examples:

- **Internet application:** Email
- **Embedded system:** Digital microwave oven
- **IoT application:** Smart home lights controlled by a phone

3. Real-World IoT Applications

What industries use IoT, and what problems does it solve?

IoT is used in healthcare, agriculture, manufacturing, transportation, and smart homes. It can help with monitoring, automation, tracking, and improving efficiency.

Choose one IoT application and explain how it works.

One example is smart farming. Sensors can check the moisture of the soil. The information is sent to a system that checks if the soil is too dry. If needed, the system can turn on the irrigation system. This helps farmers save time and water.

What are the benefits and challenges of IoT?

IoT can make work easier, save time, improve efficiency, and allow devices to be monitored remotely. However, it can also have problems such as security risks, privacy concerns, high costs, and Internet connection problems.

References

- National Institute of Standards and Technology (NIST). *Internet of Things (IoT).*
- IBM. *What is the Internet of Things (IoT)?*